Supplementary Material for "The Wrong Unit of Uncertainty: Simultaneous Conformal Bands for Repeated Cross-National Surveys"

The Authors

This online supplement collects secondary material referenced from the main manuscript. All content, numbers, and figures are reproduced unchanged; only figure numbers carry an "S" prefix.

S1 Proofs

This section gives full statements and proofs of the results of the main text, in the main-manuscript numbering. Each result is paired with a machine-checked contract test in the replication package that verifies its guarantee numerically. Where the source argument is a sketch or rests on an estimated constant, we say so.

Common setup. We observe K exchangeable source countries $c = 1, \dots, K$ plus a target $K{+}1$. Each country c has a latent attitude curve F_c (a CDF over a grid $t_1 < \cdots < t_T$) and a transport center μ (a leave-one-out mean or fitted center), with plug-in error curve $E_c(t) = \widehat{\theta}_t(P_c) - \theta_t(P_c)$. The latent transport score is $R_c = \max_{r,t} |F_c(t) - \mu(t)|/\sigma(t)$, the country-level (exchangeable-unit) score. F_c is not observed; a complex survey returns $\widetilde{F}_c = F_c + S_c$

with design error S_c, inducing the contaminated score $\widetilde{R}_c = R_c + \xi_c$ with $\mathbb{E}[\xi_c \mid R_c] = 0$ and $\mathrm{Var}(\xi_c \mid \cdot) = v_c^2$, where v_c^2 is estimated by a stratified-PSU / replicate-weight design bootstrap. Write $s_{\text{plug}}^2 = s_R^2 + \overline{v^2}$ and the design-to-total SD ratio $\rho = \sqrt{\overline{v^2}}/s_{\text{plug}} \in [0,1)$. Two targets are kept distinct throughout: the survey-estimate score $\widetilde{R}_{K+1}$ (T1) and the latent score R_{K+1} (T2), the object the political claim concerns.

S1.1 Theorem 1 (non-identification and necessity of design information)

Statement. Let the observed score be $\widetilde{R} = R + \xi$ with $R \perp \xi$ and ξ mean-zero. (i) The law of R—hence its $(1-\alpha)$ quantile—is not identified from the law of $\widetilde{R}$ without restrictions on the law of ξ. (ii) Among *non-randomized* bands measurable with respect to the observed scores only, no band whose radius is almost surely smaller than the plug-in radius—the $(1-\alpha)$ conformal order statistic of $|\widetilde{R}|$—by more than the conformal discreteness slack $\lceil(1-\alpha)(K{+}1)\rceil/(K{+}1) - (1-\alpha)$ is valid for the latent target on the subclass of design laws containing $\xi \equiv 0$. (The statement excludes uniform improvement; a band narrower on some realizations and wider on others is not excluded, which is why we phrase the main-text reading as "no observed-score band can beat the plug-in *uniformly* by more than the slack." A randomized band can shave exactly that slack and no more; and by Remark 1 no observed-score band—including the plug-in—is latent-valid over *all* admissible laws, so validity must be read relative to the stated subclass.)

Proof. (i) Convolution is not invertible without knowing the noise law. Take $\mathcal{L}(\widetilde{R}) = N(0,\sigma^2)$. Then $(R \sim N(0,\sigma^2),\ \xi \equiv 0)$ and, for any $0 < \tau^2 < \sigma^2$, $(R \sim N(0,\sigma^2 - \tau^2),\ \xi \sim N(0,\tau^2))$ both have $\xi \perp R$, ξ mean-zero, and induce the same $\mathcal{L}(\widetilde{R})$; their latent $(1-\alpha)$ quantiles $z_{1-\alpha}\sigma$ and $z_{1-\alpha}\sqrt{\sigma^2-\tau^2}$ differ. The same construction applies to any $\mathcal{L}(\widetilde{R})$ that is a nontrivial convolution. (ii) Under the member $\xi \equiv 0$ the latent target equals $\widetilde{R}_{K+1}$, which is exchangeable with $\{\widetilde{R}_c\}_{c \le K}$; the conformal rank identity gives

$\Pr(\widetilde{R}_{K+1} \le \widehat{q}) = \lceil(1-\alpha)(K{+}1)\rceil/(K{+}1)$ at $\widehat{q}$ the corresponding order statistic, so any non-randomized radius below the $\lceil(1-\alpha)(K{+}1)\rceil{-}1$-th order statistic of $\{|\widetilde{R}_c|\}$ has latent coverage $\le (\lceil(1-\alpha)(K{+}1)\rceil - 1)/(K{+}1) < 1-\alpha$ on this member. A band whose radius sits almost surely below the plug-in by more than the slack therefore undercovers the latent target on the $\xi \equiv 0$ law it cannot distinguish from the contaminated one; the only room below the plug-in is the discreteness slack, which randomized smoothing exhausts. Thus strict efficiency beyond that slack is attainable only by shrinking the class of admissible laws—i.e. by supplying the law of ξ (equivalently the v_c), which is exactly the design-bootstrap information. $\square$

S1.2 Theorem 2 (oracle validity)

Statement. With the countries exchangeable and the design law known: (a) the clustered band on $\{\widetilde{R}_c\}$ covers the survey-estimate target T1 exactly, $\Pr(\widetilde{R}_{K+1} \le \widehat{q}) \ge 1-\alpha$, using exchangeability only. (b) For the latent target T2 the same band satisfies, for every $u > 0$,

$$\Pr(R_{K+1} \le \widehat{q}) \;\ge\; 1-\alpha \;-\; \Pr(\xi_{K+1} < -u) \;-\; \Pr\left(\widetilde{R}_{K+1} \in (\widehat{q}-u,\ \widehat{q}]\right),$$

so the T1→T2 coverage gap is controlled by a noise-tail term and a local anti-concentration term, both of which vanish with the design-noise scale. Quantitatively, with countries independent (so that the target score is independent of $\widehat{q}$) and a density bound L for $\widetilde{R}$ near its upper quantile: optimizing u under a variance-only tail bound (Chebyshev) gives a gap of order $\rho^{2/3}$; under sub-Gaussian design-noise tails—the natural case for bootstrap-estimated design error—the optimized gap is $O(\rho\sqrt{\log(1/\rho)})$. We use independence, not mere exchangeability, for the anti-concentration step; the tail term needs neither. Equivalently, the *noise-inflated* band with radius $\widehat{q}+u$ covers the latent target at level $1-\alpha-\Pr(|\xi_{K+1}| > u)$ with no anti-concentration condition. (c) The oracle deconvolution band with the true target scale has width a factor $\sqrt{1-\rho^2} + o(1)$ of the clustered band, and under the scale-family assumption (A1) of Theorem $4'$ it is finite-sample *exact* for the latent target (Theorem $4'$(i)).

Proof. (a) By exchangeability of $\{\widetilde{R}_c\}_{c=1}^{K+1}$ the rank of $\widetilde{R}_{K+1}$ is uniform on $\{1, \ldots, K+1\}$; with $\widehat{q}$ the $\lceil(1-\alpha)(K+1)\rceil$-th order statistic, $\Pr(\widetilde{R}_{K+1} \leq \widehat{q}) = \lceil(1-\alpha)(K+1)\rceil/(K+1) \geq 1-\alpha$, finite-sample and distribution-free, using no assumption on the design-noise law. (b) On $\{R_{K+1} > \widehat{q}\}$, either $\xi_{K+1} < -u$ or $\widetilde{R}_{K+1} = R_{K+1} + \xi_{K+1} > \widehat{q} - u$. Hence $\Pr(R_{K+1} > \widehat{q}) \leq \Pr(\xi_{K+1} < -u) + \Pr(\widetilde{R}_{K+1} > \widehat{q} - u)$, and $\Pr(\widetilde{R}_{K+1} > \widehat{q} - u) \leq \alpha + \Pr(\widetilde{R}_{K+1} \in (\widehat{q} - u, \widehat{q}])$ by part (a). For the inflated band, $\{R_{K+1} > \widehat{q} + u\} \subseteq \{\widetilde{R}_{K+1} > \widehat{q}\} \cup \{|\xi_{K+1}| > u\}$ directly. (c) The oracle deconvolution scale satisfies $s_T^2 = s^2(1-\rho^2)$ by the law of total variance $\mathrm{Var}(\widetilde{R}) = \mathrm{Var}(R) + \mathbb{E}[v^2]$ under conditional mean-zero noise, giving the width factor $\sqrt{1-\rho^2} + o(1)$, the $o(1)$ as $K \to \infty$ from replacing the two laws' finite-K conformal quantiles by their population counterparts. □

Remark 1 (Symmetric noise alone is *not* sufficient for latent conservativeness). It is tempting to argue that symmetric design noise makes the observed-score band conservative for the latent target, on the grounds that a symmetric convolution is a mean-preserving spread. That inference is false without shape conditions on the latent score law: a mean-preserving spread does not imply quantile dominance at a fixed level. Counterexample: $R \in \{0, 10\}$ with masses $(0.85, 0.15)$ and $\xi = \pm 1$ with equal probability give $Q_{0.9}(R) = 10$ but $Q_{0.9}(R+\xi) = 9$ — the symmetric noise moves mass *below* the latent quantile — and the conformal band calibrated on the noisy scores undercovers the latent target (0.867 at $K = 200$, $\alpha = 0.10$, in the replication package's contract test `test_latent_target_bridge.py`). Bimodal latent score laws of this shape arise naturally when a minority of countries follows a distinct regime. This is why Theorem 2(b) is stated as an explicit two-term bound (small noise $\Rightarrow$ small gap) rather than as unconditional conservativeness, and why on real data we report the design-noise scale ($\widehat{\rho} \leq 0.22$ at the national level across all surveys examined, and ≤ 0.29 including the noisiest ESS age-band subgroups; Section S4) that makes the gap negligible rather than assuming it away. Under additional shape conditions (e.g. a unimodal latent law with $\widehat{q}$

above its mode, or a log-concave latent density) the classical dominance argument does go through, but we do not rely on it.

S1.3 Theorem 3 (exact finite-K validity of the deployed base band)

Statement. Let the error curves be $E_c(t) = \widehat{\theta}_c(t) - \mu(t)$, where the center μ satisfies the *symmetric-construction condition*: it is (i) a fixed function independent of the $K{+}1$ curves, or (ii) computed from a disjoint center fold, or (iii) computed *per unit* from that unit's own auxiliary data by a common rule (e.g. the country's own previous round, as in the deployed ESS certification), or (iv) a symmetric function of *all* $K{+}1$ curves including the target's (the surveyed-target validation mode). With the unstudentized cluster score $R_c = \max_t |E_c(t)|$ and the countries exchangeable, for *every* K the band $\widehat{B} = \text{center} \pm \widehat{q}$ with $\widehat{q}$ the $\lceil(1-\alpha)(K{+}1)\rceil$-th order statistic of $\{R_c\}$ satisfies $\Pr(F_{\text{new}} \in \widehat{B}) \geq \lceil(1-\alpha)(K{+}1)\rceil/(K{+}1) \geq 1 - \alpha$, distribution-free.

Proof. Under any of (i)–(iv) the map from the $K{+}1$ curves to the $K{+}1$ scores is symmetric in the units (conditional on the fold in case (ii)), so the scores $E_1, \dots, E_{K+1}$ are exchangeable elements of $\mathbb{R}^T$. If the score law is atomless the $\{R_c\}$ are a.s. distinct; with atoms (possible for discrete survey scales), break ties by a symmetric uniform randomization—the standard conformal device—which preserves exchangeability of the ranks, and the guarantee holds as stated (without randomization, ties can only make the band conservative, never anti-conservative, since the order statistic weakly increases). Therefore the rank of R_{K+1} is uniform on $\{1, \dots, K{+}1\}$, and with $\widehat{q} = R_{(m)}$, $m = \lceil(1 - \alpha)(K{+}1)\rceil$, $\Pr(R_{K+1} \leq \widehat{q}) = m/(K{+}1) \geq 1 - \alpha$. The simultaneous coverage event is pointwise equivalent to the scalar event: $F_{\text{new}}(t) \in \widehat{B}(t)\ \forall t \iff |E_{K+1}(t)| \leq \widehat{q}\ \forall t \iff \max_t |E_{K+1}(t)| \leq \widehat{q} \iff R_{K+1} \leq \widehat{q}$. Hence simultaneous coverage is $\geq \lceil(1-\alpha)(K{+}1)\rceil/(K{+}1)$, distribution-free. Intersecting the band with $[0,1]$ and applying the isotonic tightening $L(t) = \text{clip}_{[0,1]}(\max_{t' \leq t} \text{lo}(t'))$, $U(t) = \text{clip}_{[0,1]}(\min_{t' \geq t} \text{hi}(t'))$ only shrink the band while still containing the monotone $[0,1]$-valued truth, preserving validity. The split-modulation studentized variant (modulation computed on a disjoint fold) is exact by the identical argument conditional on the modulation fold. □

Remark 2 (Transport centers estimated from the calibration pool). A grand mean over the calibration countries, used as the center for an *unsurveyed* target, violates the symmetric-construction condition: each calibration country sits inside its own center (deviations shrunk by $(K-1)/K$) while the target sits outside, which biases $\widehat{q}$ downward — an anti-conservative $O(1/K)$ asymmetry. The deployed remedy is the leave-one-out deviation $E_c = \widehat{\theta}_c - \mathrm{mean}_{c' \neq c} \widehat{\theta}_{c'}$ (`pcb.inference.conformal_band.loo_deviations`): each calibration score then excludes its own unit exactly as the target's score excludes the target. The residual asymmetry (a pool of $K-1$ versus K units) adds an independent mean-noise term of variance larger by $O(1/K^2)$ to the calibration deviations; we *conjecture* the resulting perturbation is conservative-direction for sup-scores (larger convolution noise typically enlarging upper order statistics), but by Remark 1 convolution does not monotonically enlarge quantiles in general, so we state this as a heuristic with an $O(1/K^2)$ magnitude, not a bound. The asymmetry vanishes entirely under constructions (i)–(iv), which the headline ESS certification satisfies through its own-previous-round centers (case iii).

Modulation self-inclusion deficit (proof sketch; estimated constant). If instead one studentizes by an in-sample modulation $\widehat{s}(t) = \mathrm{SD}_c\, E_c(t)$ computed from the calibration curves themselves, the unobserved target does not share this denominator, which breaks exchangeability. A first-order expansion of the order-statistic bias in the number g of modulation slots each country contributes to gives a coverage deficit $\approx 0.382\,(g/K)$; the constant 0.382 is estimated (empirical $R^2 = 0.923$), not derived, and is flagged as such. Pooled modulation ($g = 1$) is safe; slotwise ($g = L$) undercovers by $\approx 0.382\,L/K$. This $O(g/K)$ deficit is why the deployed band is unstudentized.

Lemma 1 (Scalar-modulation invariance)**.** *If the modulation is a single in-sample* scalar $\widehat{s}$ *(the same number at every threshold and slot), the studentized band is* identical *to the unstudentized band, hence exact at any K: the scores are* $\max_t |E_c(t)|/\widehat{s}$*, their m-th order statistic* *is $M_{(m)}/\widehat{s}$ with $M_{(m)}$ the unstudentized order statistic, and the band radius $\widehat{q}\cdot\widehat{s} = M_{(m)}$ — the data-dependent scalar cancels identically. Self-inclusion deficits therefore arise* only *from non-constant modulation profiles $\widehat{s}(t)$, whose* shape *(not level) is what the unobserved target fails to share. This sharpens the note above: the $O(g/K)$ penalty is the price of profile estimation, and the deployed U0 band gives up only the profile adaptivity — the level normalization was free all along.*

S1.4 Theorem 4′ (estimated-law validity, with explicit assumptions)

The earlier draft stated this result as an order-of-magnitude proof sketch. The version below is a theorem: the price is three explicit assumptions, and the reward is that the oracle case becomes *exact* and the estimated case carries a proved, one-sided remainder. The assumptions are not decorative — the bimodal counterexample of Remark 1 shows that *some* shape restriction is necessary for any latent-target claim, and (A1) is exactly the restriction under which the deconvolution's distributional matching is an identity rather than an approximation.

Assumptions. (A1, scale family.) There exist i.i.d. random curves $W_1, \ldots, W_{K+1}$ and deterministic-given-design scales such that the observed calibration curves satisfy $\widetilde{E}_c(t) = \sqrt{s_R(t)^2 + v_c(t)^2}\, W_c(t)$ and the latent target curve satisfies $E_{K+1}(t) = s_R(t)\, W_{K+1}(t)$. This is strong, and we state its content plainly: it requires not only a common (e.g. Gaussian) shape family but that the design noise share the latent curves' *cross-threshold correlation profile*, so that noise changes only the scale of a common shape process. In real surveys the within-country sampling covariance across thresholds is governed by the design while the between-country profile is governed by substantive heterogeneity, so (A1) is an idealization—the price of finite-sample exactness. Bimodal mixtures of Remark 1 violate it. *(A2, anti-concentration.)* $M := \max_t |W(t)|$ has a density bounded by L on a neighborhood of its $(1-\alpha_{\mathrm{dec}})$ quantile q^*. *(A3, moments, boundedness, and bootstrap.)* The scores have finite fourth moment (kurtosis κ) and the error curves are uniformly bounded (automatic on the CDF scale, where $|\widetilde{E}_c(t)| \le 1$), so that empirical second moments and bootstrap replicate averages obey Bernstein-type sub-exponential tail bounds; and the size-B design bootstrap is unbiased for $v_c^2(t)$ with relative variance $\le c_v/B$. (Boundedness is what buys the logarithmic—rather than polynomial—dependence on KT below; without it, Chebyshev alone gives the same conclusion with $\log(KT)$ replaced by a polynomial factor and a correspondingly slower rate.)

Statement. Under (A1)–(A3), the always-deconvolve band — radius $\widehat{q} \cdot \widehat{s}_T(t)$ with $\widehat{q}$ the $\lceil(1-\alpha_{\text{dec}})(K+1)\rceil$-th order statistic of the studentized scores $\max_t |\widetilde{E}_c(t)|/\sqrt{\widehat{s}_T(t)^2 + \widehat{v}_c(t)^2}$ — satisfies: (i) with *oracle* scales ($\widehat{s}_T = s_R$, $\widehat{v}_c = v_c$) it is finite-sample *exact* for the latent target at every K; (ii) with estimated scales,

$$\Pr\left(F_{\text{new}} \in \widehat{B}\right) \;\ge\; 1-\alpha_{\text{dec}}-\underbrace{2Lq^*\,\delta_{K,B}}_{\text{scale error}}-\underbrace{\eta_{K,B}}_{\text{tail event}}-\underbrace{r_K}_{\text{order-stat}}, \qquad \delta_{K,B} = C_1\sqrt{\tfrac{(\kappa-1+z^2)\log K}{K}}+C_2\sqrt{\tfrac{c_v\log(KT)}{B}},$$

with $\eta_{K,B}, r_K = O(K^{-1/2})$ obtainable from the Bernstein bounds of (A3) and order-statistic concentration (C_1, C_2 depend on (κ, z, L, q^*) through those inequalities; we do not display numerical values), so $\varepsilon_{K,B} := 2Lq^*\delta_{K,B} + \eta_{K,B} + r_K \to 0$ at rate $\sqrt{\log(KT)/\min(K,B)}$; and (iii) the finite-$K$-safe guard makes the scale-error term one-sided: on the intersection of the guard's per-threshold coverage events (probability $\ge 1 - T\alpha_2$ by a union bound over the T thresholds, or $\ge 1-\alpha_2$ with a Bonferroni-adjusted guard level α_2/T), $\widehat{s}_T \le$ the oracle-consistent scale, so the band errs wide, not narrow.

Proof. (i) Under (A1) with oracle scales the studentized calibration scores are $\max_t |\widetilde{E}_c|/\sqrt{s_R^2+v_c^2} = \max_t |W_c| = M_c$, i.i.d., and the target's studentized score is $\max_t |E_{K+1}|/s_R = M_{K+1}$, an independent copy. The $K{+}1$ scores are exchangeable, the rank of M_{K+1} is uniform, and the order-statistic band is exact — the argument of Theorem 3 verbatim. (This also proves the oracle clause of Theorem 2(c).)

(ii) Define the multiplicative perturbations $\Delta_c(t) = \sqrt{\widehat{s}_T^2+\widehat{v}_c^2}/\sqrt{s_R^2+v_c^2}-1$ and $\Delta_0(t) = \widehat{s}_T(t)/s_R(t)-1$, and $\delta^* = \max\left(\max_{c,t}|\Delta_c(t)|, \max_t|\Delta_0(t)|\right)$. On $\{\delta^* \le \delta\}$ every studentized calibration score is sandwiched, $M_c/(1+\delta) \le \widehat{S}_c \le M_c/(1-\delta)$ (a uniform multiplicative

bound on the denominator survives the $\max_t$), hence $\widehat{q} \geq M_{(m)}/(1+\delta)$; and the target's statistic $\max_t s_R W_{K+1}/\widehat{s}_T \leq M_{K+1}/(1-\delta)$. A miss therefore implies $M_{K+1} > M_{(m)}(1-\delta)/(1+\delta) \geq M_{(m)}(1-2\delta)$, so

$$\Pr(\text{miss}) \;\leq\; \underbrace{\Pr\left(M_{K+1} > M_{(m)}\right)}_{\leq\alpha_{\text{dec}}\text{ (exchangeable ranks)}} \;+\; \Pr\left(M_{K+1} \in (M_{(m)}(1-2\delta),\, M_{(m)}]\right) \;+\; \Pr(\delta^* > \delta).$$

For the middle term, $M_{(m)}$ concentrates on q^* at rate $r_K = O(K^{-1/2})$ (binomial concentration of the empirical CDF at q^* together with (A2)); on that event the interval has length $\leq 2\delta q^*(1+o(1))$ and (A2) bounds its probability by $2Lq^*\delta + o(\delta)$. For the last term, decompose δ^* into (a) the variance-estimation error of s^2_{plug}, which has relative standard error $\leq \sqrt{(\kappa-1)/K}$ by the fourth-moment bound and, by the boundedness clause of (A3), a Bernstein tail $\Pr\left(|\widehat{s}^2_{\text{plug}}/s^2_{\text{plug}} - 1| > u\right) \leq 2\exp\{-cKu^2\}$ for u below a constant; (b) the guard subtraction, whose own standard error contributes the z-term with the same tail; and (c) the bootstrap error $\max_{c,t}|\widehat{v}^2_c - v^2_c|/s^2$, bounded by $\sqrt{c_v \log(KT)/B}$ via the Bernstein bound and a union bound over the KT cells. The $\log K$ and $\log(KT)$ factors inside $\delta_{K,B}$ are what make these exponential tails summable: choosing $\delta = \delta_{K,B}$ with C_1, C_2 large enough constants makes $\Pr(\delta^* > \delta_{K,B}) \leq \eta_{K,B} = O(K^{-1/2})$ (polynomial-in-$1/K$ bounds of any desired order are available by inflating C_1, C_2).

(iii) The deployed scale subtracts $[\,\overline{\widehat{v}^2} - z\ \mathrm{SE}\,]_+$, a lower confidence bound at per-threshold level $\alpha_2 = \Phi(-z)$: on the intersection of the T per-threshold coverage events (probability $\geq 1 - T\alpha_2$ by a union bound) the subtracted amount does not exceed the true design variance at any threshold, so $\widehat{s}^2_T \geq s^2_{\text{plug},0} - \overline{v^2}$ errs on the wide side of the oracle scale and the corresponding half of the Δ_0 perturbation can only *increase* coverage. The two-sided bound of (ii) therefore holds with its scale-error term effectively one-sided outside the stated event. One deployment honesty note: the shipped guard uses the per-threshold $z = 1.645$ ($\alpha_2 \approx 0.05$) *without* the Bonferroni adjustment, so at $T \approx 10$–26 the uniform union bound $1 - T\alpha_2$ is weak or vacuous; the one-sidedness should be read as per-threshold

protection at the deployed settings, and the Bonferroni-adjusted guard level α_2/T (obtained by passing $z = \Phi^{-1}(1-\alpha_2/T)$ to the shipped scale guard, whose z is a free parameter) is the configuration under which the uniform statement is nontrivial. □

Remark 3 (What (A1) rules out, and why that is the right boundary). (A1) is a distributional-matching condition: the observed and latent scores must sit in a common scale family indexed by the design variance. Remark 1's bimodal construction violates it and defeats *any* latent-target band built on observed scores, so a condition of this type is not an artifact of our proof but a feature of the problem. When (A1) is doubtful the pipeline does not rely on it: gate D (stability) and the conservative fallback do not use (A1), and at survey scale Theorem 5′′'s guarantee bypasses this theorem entirely. Machine check: `test_estimated_law_validity.py` verifies (i)-type exactness and the K/B decay of the remainder on a scale-family DGP, and the bimodal DGP's exclusion is verified by `test_latent_target_bridg`

S1.5 Theorem 5′ (safe-adaptive finite-K validity—the deployed pipeline)

The deployed validity argument uses *no conditional-on-selection step.* (An earlier draft argued that conditioning on a calibration-measurable selection event leaves each branch's marginal guarantee intact; that inference is false in general — the selection event and the conformal quantile are functions of the same calibration scores, so conditioning breaks the rank-uniformity that the marginal guarantee rests on. The architecture below removes the need for any such step.)

Lemma 2 (Free selection among nested bands)**.** *Let $\widehat{B}_1 \subseteq \widehat{B}_2$ be two bands (as random sets) with $\Pr(F_{new} \in \widehat{B}_1) \geq 1-\alpha$, and let $\widehat{J} \in \{1,2\}$ be* any *selection rule, measurable or adversarial, data-dependent or not. Then $\Pr(F_{new} \in \widehat{B}_{\widehat{J}}) \geq 1-\alpha$.*

This observation is elementary and belongs to the nested view of conformal prediction developed by Gupta et al. (2022); the general problem of selecting among conformal sets, including when selection is free and when it costs, is treated by Yang and Kuchibhotla

(2024). We claim no novelty for the lemma itself—its role here is architectural: engineering the deployed branches so that the lemma applies (unstudentized scores make the dominance pointwise, hence the nesting) is what removes the selection problem from the validity budget.

Proof. $\{F_{\text{new}} \notin \widehat{B}_{\widehat{J}}\} \subseteq \{F_{\text{new}} \notin \widehat{B}_1\}$ pointwise: if the selected band misses, then a fortiori the narrower band $\widehat{B}_1$ misses (on $\{\widehat{J} = 2\}$ because $\widehat{B}_1 \subseteq \widehat{B}_2$, on $\{\widehat{J} = 1\}$ trivially). Take probabilities. □

Statement (Theorem 5′). Deploy the selector of Algorithm 1 with the unstudentized PCB and conservative branches (scores $\max_t |E_c(t)|$ and $\max_t(|E_c(t)| + z\,\widehat{v}_c(t))$ respectively, both calibrated at level α_{anchor}) and the deconvolution branch calibrated at its own budget α_{dec}, where $(\alpha_{\text{anchor}}, \alpha_{\text{dec}}) = (\alpha, 0)$ when gate B is infeasible at the observed K (Proposition 1: $K < 94$ at the frozen constants, which covers every certification sample analyzed in this paper—ESS $K \approx 30$, LAPOP $K \approx 20$, and the WVS $\geq$ 2-wave sets at K = 59–77; the WVS *full item-coverage* sets at $K \approx$ 95–105 are feasible in principle but are not certification samples) and otherwise $\alpha_{\text{dec}} = \min\{\max(0.1\alpha,\, 3/(K{+}1)),\, \alpha/2\}$, $\alpha_{\text{anchor}} = \alpha - \alpha_{\text{dec}}$ — a function of K only (hence frozen, not data-dependent), floored at $3/(K{+}1)$ so the branch's conformal quantile is a genuine order statistic rather than the sample maximum near the feasibility boundary. Then, finite-sample and with no conditions on the selection rule,

$$\Pr\left(F_{\text{new},r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \;\geq\; \begin{cases} 1-\alpha, & K < 94 \text{ (deconvolution unreachable)}, \\ 1-\alpha-\varepsilon_{K,B}, & K \geq 94, \end{cases}$$

where $\varepsilon_{K,B}$ is the estimated-law remainder of Theorem 4 attached to the deconvolution branch only, reported at deployment through the observable diagnostic $\widehat{\delta}_{\text{UCB}}(D)$. For the survey-estimate target T1 the $K < 94$ statement is exact; for the latent target T2 it carries the additive T1→T2 gap of Theorem 2(b), negligible at the observed $\widehat{\rho} \leq 0.22$ (national level; at most 0.29 in the ESS subgroup scan).

Proof. The conservative score dominates the PCB score pointwise ($|E_c(t)| + z\widehat{v}_c(t) \geq |E_c(t)|$,

$\widehat{v} \geq 0$), so at the common level α_{anchor} the conservative order statistic is at least the PCB one and the two bands are nested: $\widehat{B}_{\text{PCB}} \subseteq \widehat{B}_{\text{con}}$. By Theorem 3 the PCB band is exact at α_{anchor}; by Lemma 2 any choice between the two nested branches retains $\Pr(\text{miss}) \leq \alpha_{\text{anchor}}$. When gate B is infeasible the selector can only ever choose between these two (Proposition 1 is a *deterministic* property of the frozen diagnostic, Section below), so the deployed guarantee is $1 - \alpha_{\text{anchor}} = 1 - \alpha$ with no remainder and no selection conditions. When gate B is feasible the deconvolution branch (not nested; narrower) is charged its own miss budget through a union bound:

$$\Pr(\text{miss}) \;=\; \Pr(\text{miss} \wedge \widehat{J} \neq \text{dec}) + \Pr(\text{miss} \wedge \widehat{J} = \text{dec}) \;\leq\; \alpha_{\text{anchor}} + \Pr(\text{miss}_{\text{dec}}) \;\leq\; \alpha_{\text{anchor}} + \alpha_{\text{dec}} + \varepsilon_{K,B},$$

the first term by the nested-selection argument restricted to $\{\widehat{J} \neq \text{dec}\}$ (the event $\{\text{miss} \wedge \widehat{J} \neq \text{dec}\}$ implies a miss of $\widehat{B}_{\text{PCB}}$), the second by Theorem 4's *marginal* guarantee for the always-deconvolve band at level α_{dec}. No term conditions on $\widehat{J}$. □

Remark 4. The gates (A–D) therefore no longer carry any validity burden: they are an efficiency-and-robustness device that decides *which* of three individually budgeted constructions to return. This is weaker machinery than the earlier conditional claim and strictly stronger guarantees: at survey scale the deployed band is exact, full stop, and the only remainder anywhere in the pipeline is Theorem 4's $\varepsilon_{K,B}$, confined to the $K \geq 94$ regime where deconvolution can activate and reported through $\widehat{\delta}_{\text{UCB}}(D)$ when it does.

S1.6 Proposition 1 (survey-scale unreachability)

Statement. The deployed selector activates deconvolution only if a *need* gate $\widehat{\rho}_{\text{LCB}} > \rho_0$ and a *reliability* gate $D \leq \tau_D$ both hold, with $D = \max_t \widehat{\text{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$ computed by the frozen formula below. Two facts bound these. *(a, algorithmic — deterministic.)* $\rho = \sqrt{\overline{v^2}}/s_{\text{plug}} \in [0, 1)$ with $s_{\text{plug}}^2 = s_R^2 + \overline{v^2}$; and the deployed diagnostic satisfies $D \geq \sqrt{2/(K-1)}$ *identically, for every possible dataset*, so gate B cannot open unless $K \geq 1 + 2/\tau_D^2$, i.e. $K \geq 94$ at the frozen $\tau_D = 0.147$. This is a property of the frozen procedure, not a statistical estimate: the deployed pipeline provably never deconvolves at cross-national-survey K, which is what Theorem 5′ uses. *(b, statistical — fourth-moment dependent.)* The interpretation of the floor as *the* sampling error of a K-cluster variance is exact under Gaussian scores and otherwise holds up to the kurtosis: the true relative SE of $\widehat{s}^2$ is $\sqrt{(\kappa-1)/K}\,(1+o(1))$ with κ the score kurtosis, larger than the Gaussian formula for heavy-tailed scores ($\kappa > 3$) and smaller for platykurtic ones ($\kappa < 3$).

Proof. (a) $\rho^2 = \overline{v^2}/(s_R^2 + \overline{v^2}) \in [0, 1)$, so the need gate can fire only when the design noise $\overline{v^2}$ is an appreciable fraction of the between-population signal s_R^2. The frozen diagnostic computes $\widehat{\mathrm{SE}}(\widehat{s}_T^2) = \sqrt{2s_{\mathrm{plug}}^4/(K-1) + (\mathrm{SD}_K(v^2)/\sqrt{K})^2}$ and, by construction of the finite-K-safe scale, $\widehat{s}_T \le s_{\mathrm{plug}}$ for every t; hence

$$D = \max_t \frac{\widehat{\mathrm{SE}}(\widehat{s}_T^2)}{\widehat{s}_T^2} \;\ge\; \frac{\sqrt{2s_{\mathrm{plug}}^4/(K-1)}}{s_{\mathrm{plug}}^2} \;=\; \sqrt{\tfrac{2}{K-1}}$$

as an algebraic identity of the deployed statistic (machine-checked in `test_prop1_floor.py`). Hence $D \le \tau_D$ forces $\sqrt{2/(K-1)} \le \tau_D$, i.e. $K \ge 1 + 2/\tau_D^2$. With the frozen gate-B constants $\widehat{\delta}_{\mathrm{UCB}}(D) = a + bD$, $a = 0.0061$, $b = 0.0943$, $\delta_{\max} = 0.02$, the threshold is $\tau_D = (\delta_{\max} - a)/b = 0.147$, giving $K \ge 1 + 2/0.147^2 \approx 94$. Repeated cross-national surveys have $K \le 33$ (ESS) and $\sim$ 20–40 (LAPOP, Afrobarometer, Eurobarometer), so the gate-B feasible region is empty at survey scale regardless of ρ: even if the need gate were passed, deconvolution stays blocked. (b) is the standard $\mathrm{Var}(\widehat{s}^2) = (\mu_4 - \sigma^4\,(K-3)/(K-1))/K$ expansion. □

Remark 5. Part (a) is what the validity theorem needs, and it is unconditional. Part (b) is what the *calibration* of $\widehat{\delta}_{\mathrm{UCB}}$ needs, and its fourth-moment dependence is a real limitation in one direction only: for platykurtic scores the frozen formula over-states the sampling error, so the gate is merely over-conservative (an efficiency cost); for heavy-tailed scores it under-states it, so $\widehat{\delta}_{\mathrm{UCB}}$ can under-budget the remainder — exactly the 0.02% extreme-tail failure mode the holdout exposed (Section S3), now confined by Theorem 5′ to the $K \geq 94$, α_{dec}-budgeted branch. The exact threshold $K \approx 94$ depends on the frozen gate-B constants (conservatively calibrated on the development grid); a less conservative $\delta_{\max}$ would lower it, but the calibration-free statement is the floor itself. The barrier is specific to the *cross-national* application: many-unit settings (subnational areas, schools, clinics) routinely have $K \geq 94$, where the reliability gate can open.

S1.7 Efficiency results AW-1–AW-3 and the open oracle inequality (AW-4)

Write the three branch half-widths at level $1-\alpha$ as $W_P = q_P\, s$ (PCB), $W_D = q_D\, s\sqrt{1-\rho^2}$ (deconvolution), and $W_C = q_C\, s$ (conservative).

AW-1 (low-ρ reduction, proven). As $\rho \to 0$, $\widehat{s}_T = s\sqrt{1-\rho^2} \to s$ and the deconvolution scores converge pointwise (at fixed K) to the PCB scores, so $q_D/q_P \to 1$ by continuity of the finite-sample quantile in the studentization, and

$$W_D/W_P = (q_D/q_P)\sqrt{1-\rho^2} = 1 - \tfrac{1}{2}\rho^2 + O(\rho^4)$$

by Taylor expansion. Below ρ_0 the selector returns PCB exactly, so $W_{\mathrm{adaptive}} = W_P$.

AW-2 (conservative dominance, conditional). For $\rho \geq \rho_0$ with stable deconvolution, $W_D/W_C = (q_D/q_C)\sqrt{1-\rho^2}$. The worst-case scores $(|E| + zv)/s$ stochastically dominate $|E|/s$, so $q_C \geq q_P$ and to first order $q_C \approx q_P(1 + z\kappa\rho)$ with $\kappa \in (0,1]$ the correlation of the $|E|$ and v orderings across the calibration set. Hence $W_D/W_C \approx \sqrt{1-\rho^2}/(1+z\kappa\rho) < 1$ for all $\rho > 0$, strictly and decreasing in ρ: the conservative fallback is a genuine upper band, never wider than the Bonferroni envelope and at least as wide as PCB, and deconvolution strictly narrows it whenever design noise is present. (Conditional on the stated first-order expansion of q_C.)

AW-3 (selection cannot inflate miscoverage, proven). Superseded by Theorem 5′: the nested-branch lemma (any selection between $B_{\mathrm{PCB}} \subseteq B_{\mathrm{con}}$ is free) plus the union bound over the α_{dec}-budgeted deconvolution branch give $\Pr(\text{target} \in B_{\widehat{J}}) \geq 1 - \alpha - \varepsilon_{K,B}\mathbf{1}\{K \geq 94\}$ with *no* appeal to conditional-on-selection coverage. (The earlier mixture argument, which asserted that each branch retains its marginal guarantee conditional on $\{\widehat{J} = j\}$, was invalid: selection event and quantile share the calibration data.)

AW-4 (excess-width oracle inequality, open). Let $W^* = \min(W_P, W_D, W_C)$. Conjecture: $W_{\mathrm{adaptive}} \leq W^* + r_{K,B}$ with $r_{K,B} = O(K^{-1/2}) + O(B^{-1/2})$ under Lipschitz width-in-ρ, the boundary zone where the target-blind ρ_0 disagrees with the oracle crossover having width $O(K^{-1/2})$ because $\widehat{\rho}$ is $\sqrt{K}$-consistent. This remains open: (i) $\rho_0 = 0.47$ is fixed from the development grid rather than derived as the oracle crossover $W_D = W_C$, and (ii) the Lipschitz constant of width-in-ρ is not established. It is stated as a candidate with a proof sketch, not a theorem.

S2 The finite-K mechanism and the development grid

Development: why the third gate is needed. On a Gaussian grid with known truth ($K \in \{30, 60, 120, 240\}$, ρ up to 1.8), the selector transitions from PCB to deconvolution at ρ_0 and to conservative at large ρ; but a naive deconvolution that omits gate B *undercovers* at small K—coverage falls to 0.75 at $K = 30$ in the transition zone—because the target scale is estimated from few countries (Figure S3). Adding the finite-K-safe scale lifts this to 0.82, and the reliability gate B lifts the deployed worst cell to 0.862 while abstaining to conservative wherever the remainder cannot be bounded. These four transition cells at 0.862–0.878 are the development evidence that *motivates* the safety gate and *calibrates* $\widehat{\delta}_{\mathrm{UCB}}$; they are not reused below.

The finite-K mechanism, isolated. A controlled Gaussian illustration (design SD = transport SD, ρ=0.9; `fig_theory_illustration.py`) makes the source of the undercoverage explicit and shows why the deployed rule is a selector rather than a deconvolver. Both the

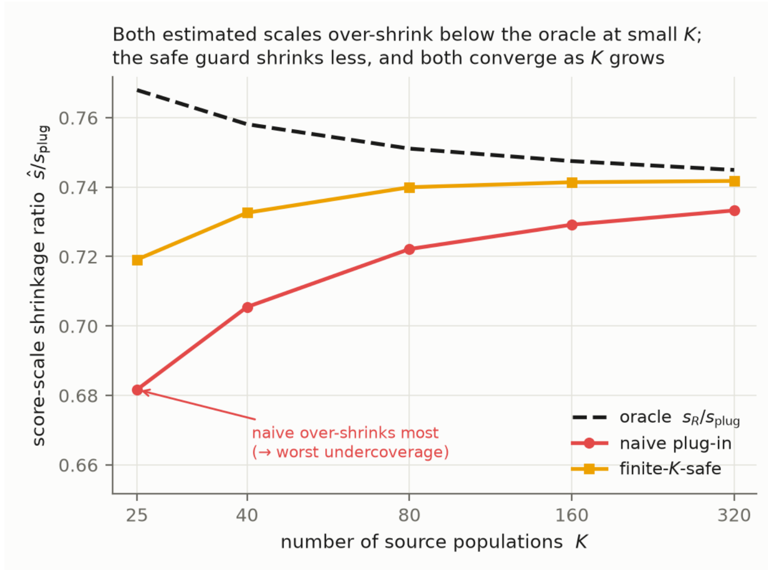

Figure S1: Score-scale shrinkage ratio $\hat{s}/s_{\text{plug}}$ vs K (Gaussian illustration, ρ=0.9, L=8). Naive and finite-K-safe deconvolution both over-shrink below the oracle at small K—the source of the finite-K undercoverage—with the safe guard shrinking less; both converge as K grows.

naive and the finite-K-safe deconvolved score scales lie *below* the oracle s_R/s_{plug} at small K (Figure S1): subtracting a design-noise variance estimated from few countries over-shrinks the scale, and the resulting band is too narrow. The safe guard shrinks less, and both converge to the oracle as $K \to \infty$. The coverage consequence is severe and grows with the trajectory length L that the band must hold simultaneously (Figure S2): naive deconvolution falls to 0.43 at K=25, L=16, and the finite-K-safe scale improves it but does not by itself reach nominal at small K. The deployed adaptive selector detects that the remainder cannot be bounded and abstains to the conservative branch, remaining valid throughout at a width cost—the mechanism behind the K=320-only activation in the confirmatory holdout in the main text.

S3 Extended robustness of the remainder bound

Corrected-scorer holdout rerun. A code audit found that the sealed holdout runner scored the PCB branch with the S2 studentized quantiles — a different band from the U0

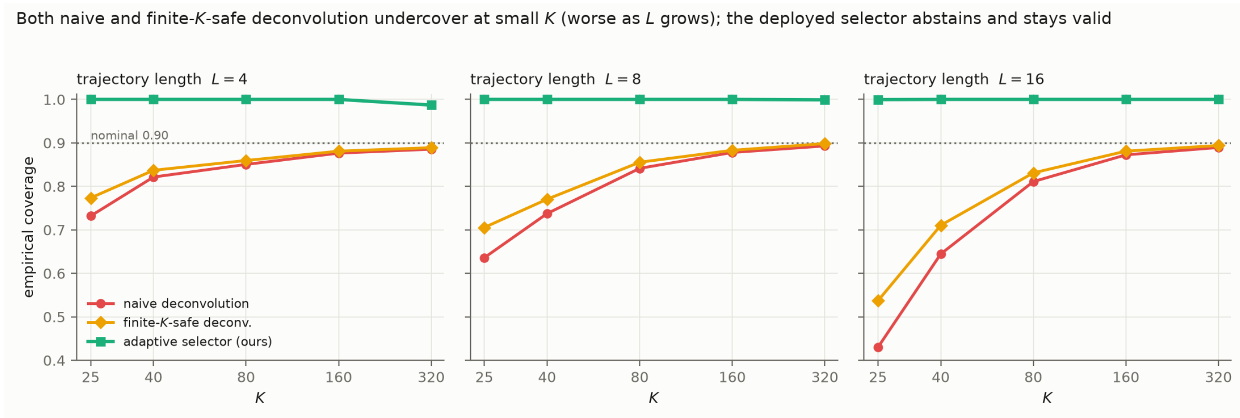

Figure S2: Latent-target coverage of the deconvolution branches vs K, faceted by trajectory length L (Gaussian illustration, ρ=0.9; nominal 0.90 dotted). Naive deconvolution undercovers severely at small K and worse as L grows; the finite-K-safe scale improves but does not itself reach nominal at small K. The deployed adaptive selector abstains to conservative and stays valid throughout.

construction the package deploys — so the sealed run certified a band the pipeline does not ship. We disclose this and report both: the sealed run (historical; it is what exposed the S2 deficit) and a full rerun of the same 450-cell grid scoring the band `dapcb` actually returns, under the revised α-budget architecture of Theorem 5$'$ and a fresh seed (the original sealed configuration file was not preserved in the archive; the grid is reconstructed exactly from the sealed run's cell table). The rerun passes every preregistered criterion outright: worst cell 0.882 (binomial-compatible with the exact level), 450/450 cells floor-compatible, deconvolution activating only at K=320, and a low-ρ width ratio of 1.005 to plain PCB.

Cell-by-cell resolution of the base-method deficit. On the holdout the studentized band's worst deployed cell was 0.843 (weak cross-round dependence at K=25), a finite-K self-inclusion deficit worst at small K and long trajectories. After switching to the unstudentized clustered band, a preregistered rerun on a fresh grid ($K \in \{15, \ldots, 60\}$, ten DGPs) confirms the fix: the worst cell rises from 0.802 (S2, K=15) to 0.893, every cell is compatible with its exact floor within the preregistered Monte-Carlo tolerance (floor minus 2 MC-SE; 29 of 120 cells sit numerically below the floor by at most 1.7 MC-SE, as expected from binomial noise

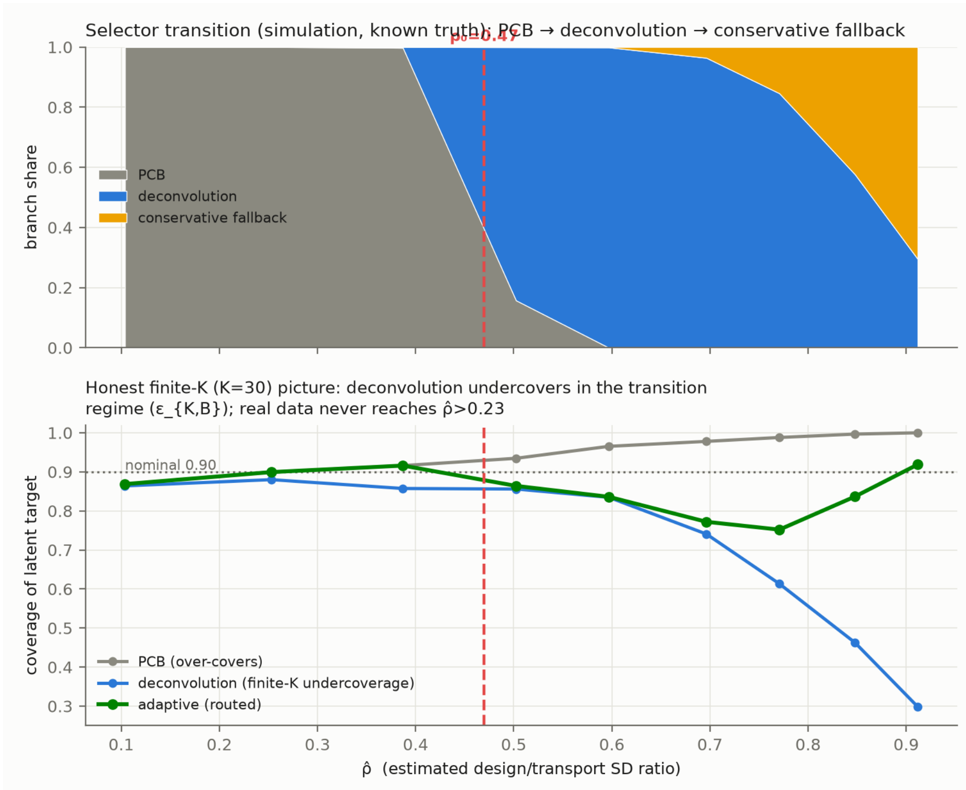

Figure S3: Development grid (Gaussian, known truth). The selector transitions PCB $\to$ deconvolution $\to$ conservative as ρ grows; naive deconvolution undercovers at small K in the transition zone—the finite-K remainder the safety gate controls.

around an exact level), and the width cost is 1.02–1.05× on the real ESS transport bands (trust-CDF errors are ≈homoskedastic across the low-trust core, so the constant-width band is competitive; the cells where it pays more than 10% are synthetic heavy-tailed and irregular-length curves at K=15—the latter at 1.17–1.18×—where the studentized alternative is itself invalid).

Extreme-tail behavior of the remainder bound. Beyond the base-method deficit resolved by deploying the unstudentized clustered band, the holdout exposed a second, extreme-tail limitation. The Gaussian-calibrated remainder bound $\widehat{\delta}_{\mathrm{UCB}}$ is not robust at the extreme: in 77 of 360,000 draws (0.02%), on strongly-dependent or heavy-tailed DGPs at K=320, draws that pass gate B undercover to ≈ 0.79. In the $\rho = 0.90$ cells the gates route 96–98% of draws to conservative and the deployed cell coverage is 0.98–0.996; at $\rho = 0.72$

Table S1: Holdout validation of the safe selector (450 cells, 800 reps each; relocated from the main text). The *sealed* run (upper rows) scored the studentized S2 base band, whose finite-K self-inclusion deficit it exposed — the finding that motivated deploying the *unstudentized* band instead. The *corrected-scorer rerun* (lower rows) scores the band the package actually returns (U0 anchors with the α-budgeted deconvolution branch) on the same grid with a fresh seed, disclosed because the originally sealed configuration file was not preserved.

criterion	target	result	status
Sealed run, S2 scorer (historical):			
P3: design-aware never causes a sub-floor cell	sub-floor $\Rightarrow$ not deconv	**all 12 sub-0.88 cells are PCB**	pass
E2: deconvolution narrower than conservative	$\leq 0.90\times$	0.577	pass
P1: cells floor-compatible (95% MC)	all 450	442/450	diagnostic
P2: worst-cell coverage	≥ 0.86	0.843	diagnostic
Corrected-scorer rerun, deployed U0/α-budget band:			
P1: cells floor-compatible (95% MC)	all 450	**450/450**	pass
P2: worst-cell coverage	≥ 0.86	**0.882**	pass
E1: low-ρ width vs PCB	$\leq 1.05\times$	1.005	pass
E2: deconvolution narrower than conservative	$\leq 0.90\times$	0.753	pass
E3: deconvolution abstains at small K	rises with K	only K=320	pass

the gates route far fewer draws away (deployed cell coverage 0.89–0.90 on the two hardest families), so the protection is concentration-dependent, not uniform over the $\rho \geq 0.72$ range. Under the revised Theorem 5′ this entire failure mode is confined to the $K \geq 94$, α_{dec}-budgeted branch and cannot touch the survey-scale guarantee.

S4 Additional empirical robustness

Design-preserving stress test, reinterpreted. Subsampling the real AmericasBarometer while preserving its stratified-PSU structure, pooled $\widehat{\rho}$ rises to ≈ 0.23 and then *turns over* at the deepest subsampling ($f = 1/16$), where the protocol's own $1/\sqrt{f}$ scaling predicts $\widehat{\rho} \approx 0.49$—i.e., a crossing of ρ_0. The discrepancy is diagnostic: at $f = 1/16$ roughly one PSU remains per stratum, where the m-of-m bootstrap (no Rao–Wu–Yue rescaling; singleton strata contributing zero) collapses the design-variance estimate exactly as the true design noise peaks, while s_{plug} continues to absorb the realized noise. What the sweep demonstrates is therefore a failure mode of the $\widehat{\rho}$ diagnostic under sparse-PSU designs—with the deployed band's coverage remaining at or above nominal throughout—not a substantive saturation law, and we have removed "saturation" claims resting on it from the main text. A rescaled-

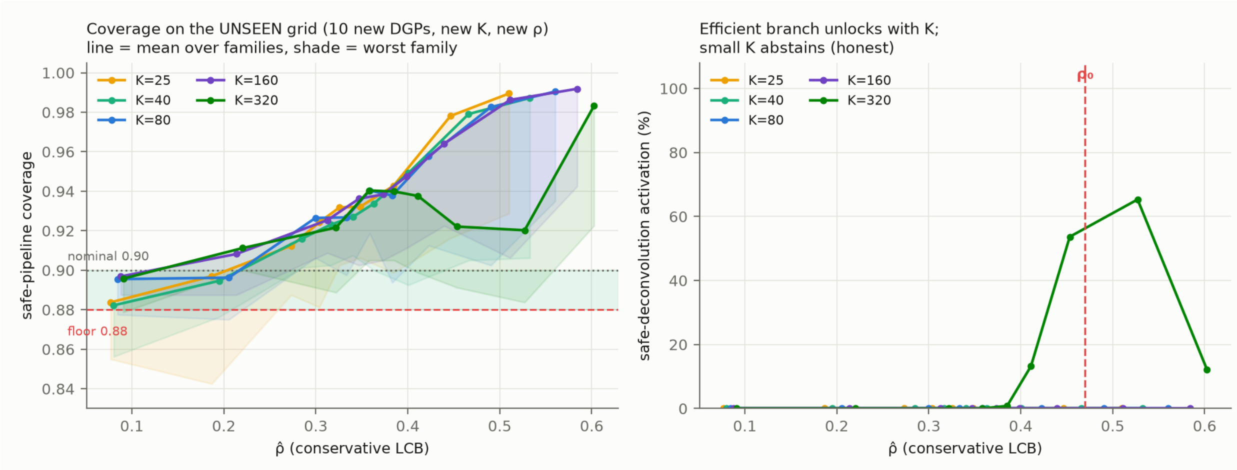

Figure S4: Holdout validation on the unseen grid (relocated from the main text). Left: deployed coverage vs. $\widehat{\rho}$ per K (line = mean over families, shade = worst family) against the 0.88 floor and 0.90 nominal. Right: deconvolution activates only at K=320; small K abstains.

bootstrap rerun (`psu_bootstrap(rescale=True)`, added to the package) is flagged for the archived replication.

The positive regime, demonstrated (E31). The scope result says where the correction is unreachable; a companion simulation shows where it pays. On the MRP-style small-area DGP (latent area curves plus known heteroskedastic posterior SDs), the frozen pipeline — unchanged constants, the α-budget architecture of Theorem 5′ — is run at $K \in \{60, 94, 150, 220, 300\}$ areas and posterior-noise ratios 0.3–0.9, scoring the returned band against the latent area curve (800 reps per cell). Three facts emerge. First, gate-B *feasibility* ($K \geq 94$) is necessary but not sufficient: with the deconvolution branch honestly budgeted at $\alpha_{\text{dec}} = \max(0.1\alpha, 3/(K+1))$, its conformal quantile is informative — and the width-gain gate clears — only from $K \approx 220$, so the practical activation frontier sits near $K \approx$ 200–300. Second, where it activates (noise ratio $\approx$ 0.7: 31% of draws at K=220, 98% at K=300) it pays: the deployed band is 0.69–0.74× the conservative envelope's width at coverage 0.98–1.00, against the guarantee floor 0.88. Third, at noise ratios near or above the between-area signal the stability gate abstains to conservative — correctly, since there the deconvolved scale hits its floor. Many-unit clustered settings (US counties, schools, clinics) have K well beyond this frontier; repeated cross-national surveys never do (`e31_positive_regime`; `results/positive_regime.csv`).

The barriers are not survey artifacts. Both gates are distribution-free, so the boundary should transfer to any conformal procedure calibrated on estimated objects (the general remark of the main text). A simulation confirms it outside the survey setting (`e29_beyond_surveys`): across Gaussian, heavy-tailed, skewed, and a non-survey model-based small-area (MRP-style) calibration, the reliability floor $D \geq \sqrt{2/(K-1)}$ holds in every case (minimum D/floor ≥ 1.06), and the gate opens only at $K \approx 130$–150—if anything a stronger barrier than the floor-implied $K \geq 94$. The one substantive difference is instructive: when a small area's posterior noise is comparable to the between-area signal ($\widehat{\rho}_{\text{LCB}} = 0.54$ at a noise-to-signal ratio of 0.5), the *need* gate does fire, unlike any cross-national survey. A many-area MRP application with appreciable posterior uncertainty is therefore the setting where the deconvolution could actually activate—the positive-result frontier the survey-scale impossibility leaves open.

The AmericasBarometer's design metadata, alongside the ESS's. The ESS publishes sample-design files (PSU, stratum, inclusion probabilities) for all rounds—integrated into the main file from round 9, as separate SDDF files earlier—so our restriction to rounds 9–11 reflects the integrated extract we analyzed, not an ESS limitation (an earlier draft misstated this); the AmericasBarometer releases full stratified-PSU structure in its merged file, letting us compute a proper design bootstrap directly. The design effect is real: the ratio of the proper stratified-PSU standard deviation to a naive independent-resampling one has median ≈ 1.13–1.20 and reaches 1.9 in the most clustered country-years. The proper band captures clustering variance the naive bootstrap misses. Yet certification is coarse-robust: the design effect moves band *width* but rarely flips the binary decline decision, because the decision is dominated by the between-country transport spread, not the within-country

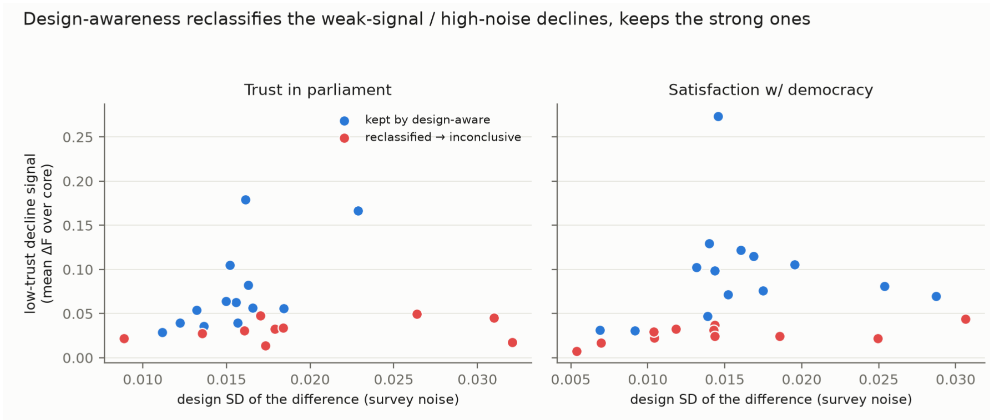

Figure S5: Relocated from the main text: countries certified by the marginal plug-in but inconclusive under the design-aware simultaneous band are the weak-signal, higher-noise cases; the apparent trend lies within the design uncertainty the survey warrants.

design noise—the same low-ρ fact, seen from the certification side.

Robustness across age groups (ESS). A preregistered age-group analysis (preregistered in the replication package) reruns the same certification within fixed age bands, changing nothing else. The "only Greece" result is not an artifact of pooling ages: it holds within the older (50+) group for both outcomes and within the middle (30–49) group for trust in parliament, so the persistent decline is concentrated in the older and middle adult population rather than created by averaging. Youth (18–29) certify *zero* persistent declines, but we read this as an underpowered null rather than youth stability: only 17 of 30 countries clear the preregistered minimum-sample gate for a single youth band, and the smaller, noisier youth cells yield wider design-aware bands that certify less—the small-K scope condition of the simulation appearing on real data. The analysis thus supports the headline while adding an honest caveat, and it is emphatically *not* a story of youth-led democratic disaffection.

Youth deconsolidation on the World Values Survey. The youth claim in the Foa–Mounk reanalysis is only half-supported: the young show *more* persistent decline in rating democracy essential (7 vs. 3 countries) but *not* in openness to strongman or army rule (0–1 vs. 4), against the "youth embrace authoritarianism" reading.

V-Dem cross-tabs of the certified core (public data; descriptive). Two checks against the V-Dem country–year data (v15; Regimes-of-the-World classification and electoral-democracy index; `e35`, results in `vdem_regime_crosstab.csv` and `vdem_predictive.csv`). *Regime type.* Of the 13 certified-core countries, five (Azerbaijan, Iraq, Lebanon, Rwanda, Uzbekistan) are *electoral autocracies* and none is a closed autocracy in the 2019 classification; five are electoral democracies (Albania, Bosnia and Herzegovina, Ecuador, Ghana, Tunisia) and three liberal democracies (Finland, Switzerland, Trinidad and Tobago). The main text's caution—that in the autocratic subset falling stated support reads as autocratic legitimation rather than deconsolidation in the consolidated-democracy sense—therefore applies to five of thirteen core members, not three as a name-by-name reading suggests. *Predictive validity, a null.* Among the 76 countries screened on ≥ 2 items, certified-core membership does *not* anticipate subsequent electoral-democracy decline (change in `v2x_polyarchy`, $2022 \to 2025$ and $2017 \to 2025$) better than the marginal any-pair flag: median changes are similar (-0.014 vs. -0.009 over $2022 \to 2025$), one-sided Mann–Whitney $p = 0.80$, Fisher $p = 0.89$. Persistent *attitudinal* deconsolidation and near-term *institutional* backsliding are, in this window, distinct phenomena— consistent with the contested attitudes-to-institutions mechanism in the recent panel literature cited in the main text, and a bound on how far the certified core should be read as an early-warning indicator. Both cross-tabs are descriptive aggregation on small N, reported as produced.

S5 Extended related work

Our work sits at the intersection of four literatures. First, **distribution-free conformal prediction** gives finite-sample coverage without modeling assumptions (Vovk et al., 2005; Lei et al., 2018). Because political data rarely satisfy exchangeability, we build on its relaxations: weighted conformal prediction for covariate shift (Tibshirani et al., 2019) and nonexchangeable conformal prediction, which bounds the coverage gap by the total variation between exchanged distributions (Barber et al., 2023). Gibbs and Candès (2021); Gibbs et al. (2024) adapt conformal sets to distribution shift and interpolate between marginal and conditional validity. Our "adaptive" selector differs in kind: it adapts not to temporal shift but to the *certified magnitude of survey design noise*, reducing to a fixed clustered method when that noise is negligible. Within the discipline, Randahl et al. (2026) bring conformal prediction to *Political Analysis* for the first time, calibrating bin-conditional intervals around a point forecast of conflict fatalities; our object is different in kind—finite-sample *simultaneous* trajectory bands with the country as the exchangeable unit and estimated calibration distributions, rather than bin-conditional intervals on a single prediction.

Second, we treat the **population (country) as the exchangeable unit**, following hierarchical conformal prediction for two-layer models (Dunn et al., 2023) and distribution-free inference with hierarchical data (Lee et al., 2023). The same finite-sample-exact clustered construction is developed, for a different (non-survey) domain, in a companion paper by the present authors (reference omitted for anonymous review). These methods transport guarantees to an unseen cluster—our cross-national setting—but treat each cluster's data as cleanly sampled. Our departure is that the calibration objects are not observed: each country's attitude distribution is *estimated* under a complex survey design, so design-based sampling variance (Binder, 1983; Lumley, 2010) contaminates the very scores conformal calibration relies upon. The nested view of conformal prediction and selection among conformal sets are developed by Gupta et al. (2022) and Yang and Kuchibhotla (2024); our Lemma 2 is an instance of that machinery, and our contribution is architectural (engineering the branches so the lemma applies), not the lemma itself.

Third, the closest methodological neighbors handle **noisy or estimated calibration**. Wieczorek (2023) develops design-based conformal prediction under complex sampling, but design enters through test-point weighting rather than as noise in transported objects. Einbinder et al. (2024) and Sesia et al. (2025) correct conformal calibration under label noise, the latter by deconvolving a *known* noise model. Our central theoretical result is that this identification premise generally fails for survey design noise: recovering the latent transport-error law is a deconvolution problem (Fan, 1991) that is non-identified without a design-noise law, so no observed-score band can beat the conservative plug-in by more than the discreteness slack while remaining valid. And even when a noise model *is* supplied, the deconvolution's reliability is itself bounded at a finite number of calibration units by an algorithmic floor—a finite-K obstruction distinct from the asymptotic identifiability the classical deconvolution literature studies.

Fourth, our target is a **functional object**—a CDF over thresholds tracked across rounds—placing us near simultaneous functional conformal bands (Diquigiovanni et al., 2021). Procedures like FDR or Bonferroni (Benjamini and Hochberg, 1995) adjust a collection of tests after the fact; our guarantee is instead carried by the geometry of a single finite-sample band, so the "persistent" object is *defined and covered directly*, not recovered by correcting marginal p-values. The distinction is one of estimand, not merely error rate, and partial-pooling cautions about reading marginal country trends (Gelman et al., 2012) motivate but do not deliver the stronger claim. Claassen-style latent-variable panels (Claassen, 2019, 2020) deliver model-based credible intervals for a latent mood under a dynamic measurement model, pooling across items and years; our bands deliver finite-sample, model-free coverage for a distribution-level object at a stated claim rung, at the price of wider intervals and no pooling. The two are complements—a certification that survives our rung hierarchy and falls inside a Claassen-style interval is doubly supported—and a same-data comparison is a natural companion exercise.

References

Barber, R. F., E. J. Candès, A. Ramdas, and R. J. Tibshirani (2023). Conformal prediction beyond exchangeability. *The Annals of Statistics 51*(2), 816–845.

Benjamini, Y. and Y. Hochberg (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B 57*(1), 289–300.

Binder, D. A. (1983). On the variances of asymptotically normal estimators from complex surveys. *International Statistical Review 51*(3), 279–292.

Claassen, C. (2019). Estimating smooth country-year panels of public opinion. *Political Analysis 27*(1), 1–20.

Claassen, C. (2020). Does public support help democracy survive? *American Journal of Political Science 64*(1), 118–134.

Diquigiovanni, J., M. Fontana, and S. Vantini (2021). Conformal prediction bands for multivariate functional data. *arXiv preprint arXiv:2106.01792*.

Dunn, R., L. Wasserman, and A. Ramdas (2023). Distribution-free prediction sets for two-layer hierarchical models. *Journal of the American Statistical Association 118*(544), 2491–2502.

Einbinder, B.-S., S. Bates, A. N. Angelopoulos, A. Gendler, and Y. Romano (2024). Label-noise robustness of conformal prediction. *Journal of Machine Learning Research 25*. arXiv:2209.14295.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *The Annals of Statistics 19*(3), 1257–1272.

Gelman, A., J. Hill, and M. Yajima (2012). Why we (usually) don't have to worry about multiple comparisons. *Journal of Research on Educational Effectiveness 5*(2), 189–211.

Gibbs, I. and E. Candès (2021). Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, Volume 34.

Gibbs, I., J. J. Cheng, and E. Candès (2024). Conformal prediction with conditional guarantees. *Journal of the Royal Statistical Society Series B*. arXiv:2305.12616.

Gupta, C., A. K. Kuchibhotla, and A. Ramdas (2022). Nested conformal prediction and quantile out-of-bag ensemble methods. *Pattern Recognition 127*, 108496.

Lee, Y., R. F. Barber, and R. Willett (2023). Distribution-free inference with hierarchical data. *arXiv preprint arXiv:2306.06342*. Forthcoming, ACM/IMS Journal of Data Science.

Lei, J., M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association 113*(523), 1094–1111.

Lumley, T. (2010). *Complex Surveys: A Guide to Analysis Using R.* Wiley.

Randahl, D., J. P. Williams, and H. Hegre (2026). Bin-conditional conformal prediction of fatalities from armed conflict. *Political Analysis 34*(1), 96–108.

Sesia, M., Y. X. R. Wang, and X. Tong (2025). Adaptive conformal classification with noisy labels. *Journal of the Royal Statistical Society Series B 87*(3), 796–815.

Tibshirani, R. J., R. F. Barber, E. Candès, and A. Ramdas (2019). Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, Volume 32.

Vovk, V., A. Gammerman, and G. Shafer (2005). *Algorithmic Learning in a Random World.* Springer.

Wieczorek, J. (2023). Design-based conformal prediction. *Survey Methodology 49*(2). Statistics Canada, Catalogue No. 12-001-X. arXiv:2303.01422.

Yang, Y. and A. K. Kuchibhotla (2024). Selection and aggregation of conformal prediction sets. *Journal of the American Statistical Association 119*(545), 435–447.